Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

Marcus E. Madsen*
Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA

Audrey K. Christiansen*
Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA

Jinkwan Han*
Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA

Jeffrey R. Hill†‡[1]
Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA
email: jeff.hill@byu.edu

Sterling G. Baird†‡[1]
Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA
email: sterling.baird@byu.edu

Tensegrity-inspired rigid–soft architectures are candidates for impact mitigation, but their transmitted-shock response is difficult to predict from simulation alone. We developed an experiment-driven Bayesian optimization framework for a co-printed polylactic-acid/thermoplastic-polyurethane three-strut-prism family with five continuous geometry variables. Candidate geometries were projected to a common mass budget, fabricated in single multi-material builds, and characterized by 101 repeated instrumented drops each; the optimizer jointly minimizes the filtered peak-acceleration ratio transmitted to the payload and a mass-weighted rebound score, with simulation restricted to a screening role. In the nine-design Sobol seed round the peak-acceleration ratio spanned 0.893 to 1.062: five articles amplified the peak shock and one attenuated it. A fully Bayesian sparse-prior surrogate with noisy expected hypervolume improvement then recommended nine designs, printed and tested under the identical protocol (round-2 values in this draft are clearly marked synthetic placeholders pending those tests). The best recommended design reduced the ratio to 0.858, eight of nine measured below unity, and the dominated hypervolume grew 79%, versus 19% for a budget-matched quasi-random baseline. The workflow transfers to other experimentally evaluated architected structures whose constitutive response resists first-principles calibration.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, impact mitigation, design of experiments

Draft note (remove before submission): this draft is written at the intended final scope of the study. Every number printed in this violet color, in particular the round-2 "measured" values of Table 6 and Figs. 9 to 11, is *synthetic placeholder (dummy) data*, invented so the paper reads end-to-end; it will be replaced verbatim by the campaign pipeline once the printed round-2 batch completes its drop sessions. Quantities in ordinary black type are real measurements. The quasi-static, replicate, and metal-analog studies remain explicit placeholders.

1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property, that no two rigid bars touch, yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1, 2]. These properties have made tensegrity attractive across scales, from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers. Throughout, we use *tensegrity-inspired* for printable cells that borrow the separated compression-member and tension-network geometry of tensegrity while relaxing defining mechanical conditions of classical tensegrity. In the present specimens, the TPU tension elements are extensible, the PLA struts may share printed junctions, and no pretension is activated or measured. We therefore do not claim classical tensegrity, prestress-stabilized behavior, or equivalence to an ideal cable–strut prism.

Multi-material fused-deposition modeling (FDM) that pairs a rigid polymer such as polylactic acid (PLA) with a soft, rate-dependent elastomer such as thermoplastic polyurethane (TPU) has been used to fabricate rigid–soft architected energy absorbers on a single platform: thick-panel origami metamaterials with PLA panels and TPU hinges [12] and ABS/TPU honeycombs with tunable stiff/flexible proportions [13]. The same rigid–soft co-printing capability makes it possible to rapidly fabricate diverse tensegrity-inspired geometries, which adopt a discrete-compression/continuous-tension topology rather than an origami or honeycomb one. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed and tested; and the relevant impact-performance metrics (the shock transmitted across the structure and the energy returned to the protected payload) are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17, 18]. The gap this paper addresses is the combination: Pajunen et al. [11] printed tensegrity-inspired cells under a single fabrication condition with no optimization loop, and Mo et al. [17] closed a multifidelity

[1]Corresponding Author.
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BO loop entirely in simulation. Among the studies reviewed here, prior work separately demonstrates physical impact testing of printed tensegrity-inspired structures, rigid–flexible additive fabrication, and Bayesian optimization of architected materials; we found no prior report combining these elements in a BO campaign over co-printed multi-material tensegrity-inspired hardware with physical impact measurements as the optimization ground truth. The campaign reported here closes the full recommend–fabricate–measure–update cycle: the Sobol seed round trains the surrogate, the model-recommended batch is fabricated and tested under the identical protocol, and the surrogate is refit on the pooled data (the round-2 measured values in this draft are the clearly marked synthetic placeholders described in the draft note above).

We emphasize one framing point at the outset: the printed PLA/TPU T_3 prism studied here is a testbed for developing and validating the closed-loop workflow, not flight hardware. The planetary-lander setting motivates the objectives (limit the shock delivered to a payload, limit the energy returned to it) and the constant-mass fairness constraint, but the contribution is the loop itself, which is agnostic to the eventual application scale.

Contributions. This paper makes the following contributions:

(1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression members and TPU tension elements that are anchored *inside* the ends of each PLA strut (the strut acting as a rigid cage in which the cables meeting at a given end join before exiting through discrete outlets). This study reports the resulting printable joint geometry; pull-out strength, fatigue life, and any durability advantage over exposed interfaces remain untested. Candidate designs are compared fairly through a constant-mass projection that rescales every geometry to a common material budget before printing (Section 3.1).

(2) A characterized drop-tower protocol and objective stack for printed impact absorbers: 101 repeated instrumented drops per specimen, standardized SAE J211 filtering [19], a velocity-change rig-health gate, and two per-specimen objectives (the filtered peak-acceleration ratio across the specimen and a mass-weighted rebound score), each characterized for repeatability before entering the optimizer (Section 3.3).

(3) An experiment-driven BO loop, built on BoTorch/Ax [20, 21] with a fully Bayesian sparse-prior surrogate (SAASBO) [22], that ingests the physical drop measurements, models per-print mass explicitly, and recommends the next batch of designs to fabricate; both completed rounds (the Sobol seed round and the first model-recommended batch) are reported here (Sections 3.4 and 4).

(4) A physics-based simulation ladder (rigid-body through finite-element tiers) used for screening and hypothesis generation alongside the physical loop, with preliminary agreement and documented blind spots assessed against the measured seed round (Section 3.5).

(5) A planned exploratory metal-analog comparison (hollow aluminum struts with threaded stainless-steel cables) that probes whether the learned design ranking transfers across material systems, spanning predicted worst, mediocre, and best designs rather than only the optimum; construction, mass-matching, and replication details remain to be fixed before the comparison can distinguish geometric rank transfer from material-system confounds (Section 3.6).

The completed evidence in this paper is restricted to single-cell, fixed-topology T_3-prism-inspired specimens under one *axial* drop condition; the planned quasi-static extension is reported separately as pending. Off-axis and combined loading, controlled fatigue and damage-accumulation testing, multi-cell behavior, and landing-attitude variability are outside the present evidence base and are identified as future work (Section 5); the novelty and generality claims below are correspondingly scoped to this restricted prototype design space rather than to tensegrity-inspired absorbers in general.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3 introduces the design parameterization, fabrication workflow, drop-tower protocol and objectives, BO loop, simulation screening ladder, and metal-analog validation plan. Section 4 reports both campaign rounds, the surrogate audits, and the optimization gain over a budget-matched baseline, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. Santos [23] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ∼23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [24–26]. Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites. More recent work on dynamically loaded additively-manufactured energy-absorbing lattices [27], together with Intrigila et al.'s bistable tensegrity-like unit cell for lattice metamaterials [28] as the closest published analog, confirms that this architecture class also responds favorably under impact and quasi-static loading, although neither study employed multi-material FDM or closed-loop design optimization. Table 1 situates the present work against selected prior efforts spanning the three elements combined here (impact-tested printed tensegrity-inspired structures, rigid–flexible fabrication, and BO of architected materials), making explicit that, among the studies reviewed, no single prior study combines a multi-material FDM tensegrity-inspired architecture with an *experiment*-driven optimization loop.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid–soft FDM enables stiff/compliant combinations on a single platform with tunable energy absorption, demonstrated for thick-panel origami with PLA (or ABS/CFRP) panels and TPU hinges [12] and for ABS–TPU honeycombs [13]. Neither study is itself a tensegrity, but both establish the rigid–soft co-printing route we adopt. Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, reported to reduce interface delamination under cyclic loading, something we take inspiration from in our designs. Beyond these rigid–soft demonstrations, a growing body of work confirms that genuinely tensegrity-inspired architectures can be additively manufactured and exhibit programmable mechanical response. Bauer et al. [32] showed that 3D-printed tensegrity metamaterials delocalize deformation to resist catastrophic failure, while Pajunen et al. [33] tuned acoustic bandgaps in 3D-printed tensegrity-inspired lattices through prestrain, and Sabouni-Zawadzka et al. [34] experimentally characterized 3D-printed tensegrity-inspired metamaterials built on the simplex module. More directly relevant to impact, Almeida et al. [29] and Davami et al. [30] characterized the high-strain-rate and dynamic response of additively manufactured tensegrity-inspired structures, and Wang et al. [31] reported an integrated fabrication-and-validation workflow for tensegrity-inspired rigid–flexible metamaterials. Collectively these studies establish the

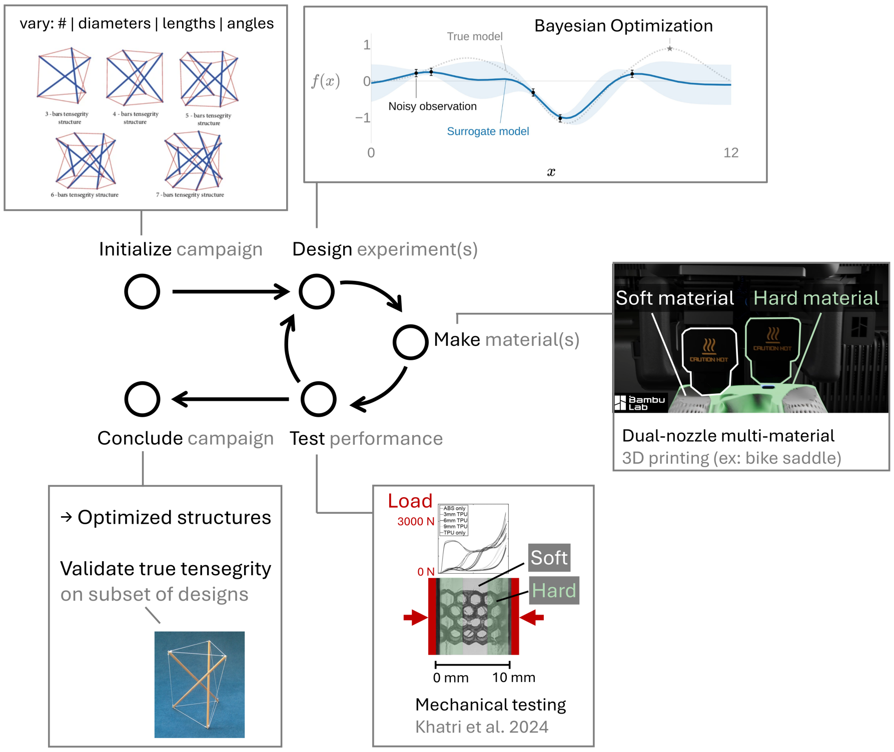

Fig. 1 Closed-loop, experiment-driven design framework. Parameterized PLA/TPU tensegrity-inspired unit cells are printed, characterized by repeated instrumented drop-weight impacts, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs to fabricate.

additive-manufacturability, tunability, and impact relevance of the architecture class adopted here, none combining multi-material FDM with an experiment-driven optimization loop. Recent multi-material PLA/TPU sandwich and layered composites [35] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while FDM PLA fatigue data [36, 37] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [38]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [39, 40].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [41, 42]. Modern implementations such as BoTorch [20] and the Ax platform built atop it [21] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [43], noisy expected hypervolume improvement for multi-objective settings [44], input-noise-robust acquisitions [45], and evolution-guided constrained multi-objective BO for self-driving labs [46]. Mixed/categorical search spaces, needed for the planned connectivity-topology extension rather than the continuous T_3-prism prototype studied here, have dedicated treatments in BOCS [47] and the one-hot-with-rounded-kernel construction of [48]. BO has been applied to architected-material design [17] and to additive manufacturing [18]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [49]; the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables. For the working T_3-prism prototype that defines the initial BO search space (Section 3.4), only the first two groups are active and all of their variables are *continuous*; the connectivity-topology and tiling groups are held fixed for this prototype and are listed here as planned extensions of the design family:

- **PLA compression members.** Strut diameter d_s and length ℓ_s, with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections).
- **Connectivity topology (future extension).** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); when activated, the design vector would encode this as a categorical choice. It is held fixed at the T_3 prism for the present campaign.
- **Unit-cell tiling (future extension).** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction; held fixed at a single cell here.

Working prototype. The first instance of this family used to validate the fabrication and test pipeline is a three-strut, T_3-prism-inspired cell scaled to a ~50 mm bounding box. As implemented in the BO harness, this working prototype exposes five continuous design variables (base radius R, cell height H, twist angle θ, a single strut diameter d_s, and a single cable (tendon) diameter d_t) over the ranges listed in Table 2; the cable diameter d_t is treated as a *continuous* variable on $[3.0, 5.5]$ mm rather than a discrete set. Under the prism's D_3 symmetry all struts share one diameter

Table 1 Positioning of this work against selected prior efforts spanning the elements combined here. The present campaign uses physical drop measurements of peak-shock transmission (t_{180}) and a provisional mass-weighted rebound score (E_{reb}) as its BO responses; it does not optimize specific energy absorption or crush energy.

Study	Architecture	Fabrication	Optimization	Response optimized	Ground truth
Pajunen et al. 2019 [11]	Truncated-tetrahedral tensegrity-inspired	Single-material 3D print	None (single condition)	n/a	Experiment (drop impact)
Intrigila et al. 2022 [28]	Bistable tensegrity-like lattice unit cell	Single-material 3D print	None	n/a	Experiment (quasi-static compr.)
Mo et al. 2023 [17]	Architected lattices	n/a (simulation)	Multifidelity BO	Simulated response metric	Simulation (+ sparse high-fid.)
Almeida 2025 [29]; Davami 2025 [30]	Tensegrity-inspired structures	Additive manufacturing	None reported	n/a	Experiment (high-rate/dynamic)
Wang et al. 2026 [31]	Tensegrity-inspired rigid–flexible metamat.	Integrated dual-material	None reported	n/a	Physical validation
This work	Tensegrity-inspired T_3-prism cell (PLA/TPU)	Multi-material FDM (PLA + TPU)	Experiment-driven BO (SAASBO/qNEHVI)	t_{180}; provisional E_{reb}	Physical experiment

and all cables share one diameter, so the present parameterization uses a single strut-diameter axis and a single cable-diameter axis (two member-diameter axes), leaving room to relax to per-orbit (saddle, top, bottom) or fully per-member diameters once the individual orbits have been characterized. The joint diameter d_j is held fixed at 7.0 mm rather than optimized: it sets the printed strut-end cage that houses the internal cable anchors (Section 3.2), so it is constrained primarily by the TPU-tendon outlet count and the dual-nozzle clearance rather than by energy-absorption performance, and fixing it keeps the anchor geometry constant across specimens so that the five optimized variables are not confounded by joint-strength changes; relaxing d_j is deferred to a follow-on joint-design study. Figure 2 shows the parametric CAD model alongside an as-printed prototype of this working T_3 prism.

Constant-mass projection and printability screens. Because cell mass varies severalfold across the raw design box, an unconstrained search would partly reward designs for carrying more material rather than for their architecture. Every candidate is therefore projected onto a constant material budget before printing: the design is uniformly rescaled (all dimensions except the twist angle) until its solid CAD mass $\rho_{PLA}V_{PLA} + \rho_{TPU}V_{TPU}$ equals $m^{\star} = 30.95$ g, the solid mass of the reference prism, converged to within 0.15 g on rendered geometry. Two printability screens are evaluated on the projected geometry and recorded as flags rather than hard rejections: a cylindrical build envelope $\pi R^2 H \leq 250\,\mathrm{cm}^3$ and the 3.0 mm TPU self-bridging floor on the as-projected cable diameter (the projection can scale a nominally feasible cable below the floor; the two seed designs where this occurred both exhibited tendon stringing defects, consistent with the screen). The choice of a *solid-mass* budget has a documented limitation discussed in Section 5: PLA struts print at partial infill while thin TPU cables print near solid, so as-printed masses still ranged from 18.50 g to 22.29 g across the seed batch; a calibrated printed-mass model that closes this gap is described in the Supplementary Information.

As-built resolution of the continuous variables. Although d_s and d_t are treated as continuous, the as-built cross-sections are discretized by the nozzle diameter, the slicer line width, and support interaction, so the effective search granularity is coarser than the nominal continuous box. The realized member diameter therefore changes in steps set by the line-width quantization rather than continuously, and the smallest CAD change that yields a reliably distinguishable as-built member is bounded below by these slicer limits; the lower TPU-tendon bound (3.0 mm) is itself a workflow-specific limit tied to the TPU self-bridging and auto-support detection floor on this H2D platform rather than a fundamental geometric one. The as-built quantity measured for every article is its mass, weighed on a 0.01 g scale; dimensional metrology of the printed members has not yet been performed, so the geometry attached to the surrogate is the commanded post-projection geometry, and

Table 2 Design variables for the T_3-prism campaign, as implemented in the BO harness. The seed round is an $n = 9$ Sobol draw (fixed seed) over this box, with each design then projected onto the constant solid-mass budget $m^{\star} = 30.95$ g. The joint diameter is held fixed; all specimens are printed in the vertical orientation on a Bambu Lab H2D, one specimen per named build plate.

Design variable	Symbol	Units	Range
Base radius	R	mm	25–40
Cell height	H	mm	60–110
Twist angle	θ	deg	40–80
Strut (PLA) diameter	d_s	mm	6.0–12.0
Cable (TPU) diameter	d_t	mm	3.0–5.5
Joint diameter (fixed)	d_j	mm	7.0

the print-to-print spread (mass standard deviation 0.457 g across the one design printed in triplicate) is carried into the observation noise model (Section 3.4).

3.2 Multi-Material Fabrication. Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. Figure 3 summarizes the end-to-end fabrication and characterization workflow, from parametric CAD through multi-material slicing, dual-extrusion printing, post-processing, and mechanical testing. Rather than wrapping the struts in a continuous TPU skin, the current design anchors the TPU tension elements *inside* the ends of each PLA strut: the cables converging at a given strut end are joined within the strut, which acts as a rigid cage housing the junction, and the individual tendons then exit through discrete outlets. This keeps the soft–rigid interface in internal, compression-dominated pockets rather than at an exposed overmolded surface. This construction effectively inverts the material roles of Ye et al.'s soft-skin-over-rigid-core wrapping [12]: here a soft TPU core is captured within a rigid PLA shell, so we cite that work only as a multi-material co-printing precedent rather than as validation of the present junction.

Print platform and joint geometry. The working platform is a Bambu Lab H2D printer, which permits a single rigid–soft build without filament swaps. Strut endpoints are tied to cables through parametric joint geometries developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants); the working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies, with a captive TPU core routed inside the PLA shell to keep the soft tendon protected from layer-line failure at the strut-to-cable transition. The full five-design comparison and the selected junction are detailed in the Supplementary

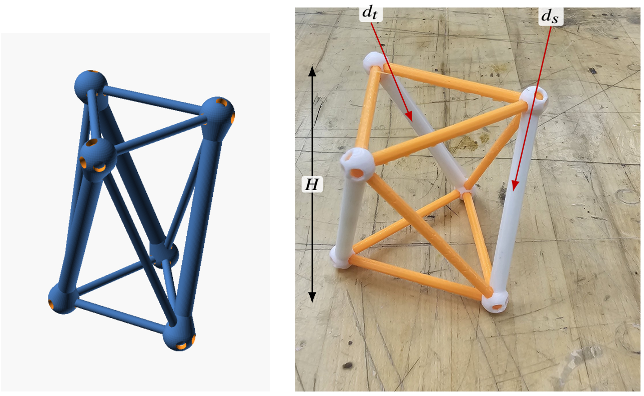

Fig. 2 Working T_3-prism prototype. ***(a)*** **Parametric OpenSCAD model whose base radius R, cell height H, twist angle θ, strut diameter d_s, and cable diameter d_t are the design variables of Table 2.** ***(b)*** **A representative dual-material print from the PR #35 print history (rigid PLA struts, here in white; tendon members in orange), with callouts marking the strut diameter d_s, a tendon (cable) diameter d_t, and the cell height H; the inter-plate twist θ is the relative rotation of the top and bottom node triangles, and the base radius R sets their circumradius.**

Information.

Supports for soft members. Because near-vertical TPU tendons are otherwise unsupported during printing, automatic support generation is disabled and supports are painted manually in the slicer: PLA edge supports run along each tendon, with the support-to-object clearance tightened from 0.35 mm to 0.2 mm during tuning on the reference prism. Two support-removal artifacts recur in the seed batch (surface stringing and, on two articles, TPU strands pulled off bottom tendons during support removal) and are logged per article in the print key (Supplementary Information). An automated alternative (a lowered support-threshold angle plus mesh-ray-cast narrowing pillars) was developed in parallel and is documented in the Supplementary Information.

Moisture and nozzle management. TPU is hygroscopic, so the soft filament is fed from a heated dry box held near 8 % relative humidity, logged per print. Midway through the seed batch the TPU extruder developed a progressive flow taper (a partial clog, compounded by a stray PLA fragment found inside the extruder); it was resolved by a cold pull and by moving the TPU to a dedicated high-flow 0.6 mm nozzle, after which print quality was restored. The affected and post-fix articles are identified in the print key so that any nozzle-related covariate can be checked against the measured objectives.

3.3 Experimental Characterization.

Drop-tower fixture. The primary characterization is repeated drop-weight impact on a Lansmont Model 23 shock test system. The printed specimen rides upright on the carriage plate, which is raised 60 in (1.524 m) along guide rails and released; the plate lands on a single 0.5 in Shore 60 OO polyurethane cushioning sheet, which shapes the landing into a single sharp base pulse of roughly 1.6 ms duration. This impact-surface configuration was selected by a blinded crossover experiment over three candidate mat arrangements, and unlike a felt stack it does not measurably compact over hundred-drop sessions. Rig health is monitored continuously through the impact velocity change Δv recovered from the base accelerometer: over an uninterrupted 100-drop characterization session the Δv coefficient of variation was 0.54% with

Table 3 Multi-material print parameters on the Bambu Lab H2D, transcribed from the archived as-printed Bambu Studio project (`t3-prism-bo-batch.H2D-MM-PLAstruts-TPUcables.as-printed.3mf`, Bambu Studio 2.7.1). Filaments are Bambu Lab PLA Basic (struts, support material) and Bambu Lab TPU 85A (tendons); the print profile is "0.30 mm Standard @BBL H2D 0.6 nozzle." The reporting structure follows the methods section of Ye et al. [12].

Parameter	PLA (struts)	TPU 85A (tendons)
Nozzle	0.6 mm	0.4 then 0.6 mm HF[a]
Nozzle temperature (°C)	220	220
Bed temperature, profile (°C)	55	35
Layer height / line width (mm)	0.30 / 0.62	
Infill	15% grid, 2 walls	near-solid
Outer / inner wall speed (mm/s)	200 / 300 (profile)	
Filament drying	n/a	dry box, ~8% RH
Build-plate type	Textured PEI	
Build orientation	Vertical	
Supports	Manual painted tree supports, PLA	

[a]The TPU side ran a standard 0.4 mm nozzle until the mid-batch clog, then a dedicated high-flow (HF) 0.6 mm nozzle; affected articles are flagged in the print key.

no drift, and every campaign session is gated on Δv remaining at the healthy tower baseline. One drop takes roughly 42 s, so a full 101-drop specimen characterization completes in about ninety minutes.

Instrumentation and signal processing. A single-axis piezoelectric accelerometer on the carriage plate records the base input and triggers the acquisition; a triaxial accelerometer (Dytran 3133A4) seated in a printed pocket at the specimen's top vertex records the transmitted response. The two base-capable channels were cross-calibrated in place over fifteen co-located drops (ratio 0.953 ± 0.001 at matched filtering), and the transducer settings are archived with the data. Signals are captured at 1.25 MHz over a 100 ms record with a 2 ms pre-trigger. All channels are low-pass filtered with the SAE J211/1 channel-frequency-class (CFC) filters [19], implemented from the standard's published coefficients with the two-pass corner frequency handled explicitly (a common implementation shortcut narrows the passband by roughly 20%, which we identified and corrected before the campaign; all reported numbers use the corrected filter). CFC-180 is the reporting class for peak metrics, with CFC-1000 retained as a higher-bandwidth diagnostic.

Per-specimen objectives. Each drop session contributes 101 valid captures to the stabilized statistics (103 for one article, autv5r), counted after the first two drops of each session are discarded as warm-up. Each session yields two objectives, each reported as a stabilized mean with the per-drop scatter quoted as one standard deviation; the standard error of the mean is what enters the surrogate's noise model (Section 3.4):

$$\begin{aligned} t_{180} &= \frac{\max_t \|\mathbf{a}_{\rm top}(t)\|_{\rm CFC180}}{\max_t |a_{\rm base}(t)|_{\rm CFC180}}, \\ E_{\rm reb} &= e_{\rm reb}\, m\, g\, h, \qquad e_{\rm reb} = \frac{g\, t_{\rm second}}{2\,\Delta v}, \end{aligned} \tag{1}$$

where t_{180} is the CFC-180 *filtered peak-acceleration ratio*: the peak magnitude of the triaxial top-vertex response over the peak base input within the impact window. We deliberately do not call this quantity a transmissibility: the two peaks need not be simultaneous and a single transient drop cannot produce the averaged frequency-response function that the standards-defined transmissibility requires, so t_{180} is a filter-class-specific ranking metric rather than a standards quantity, and SAE J211/1 is cited here only for

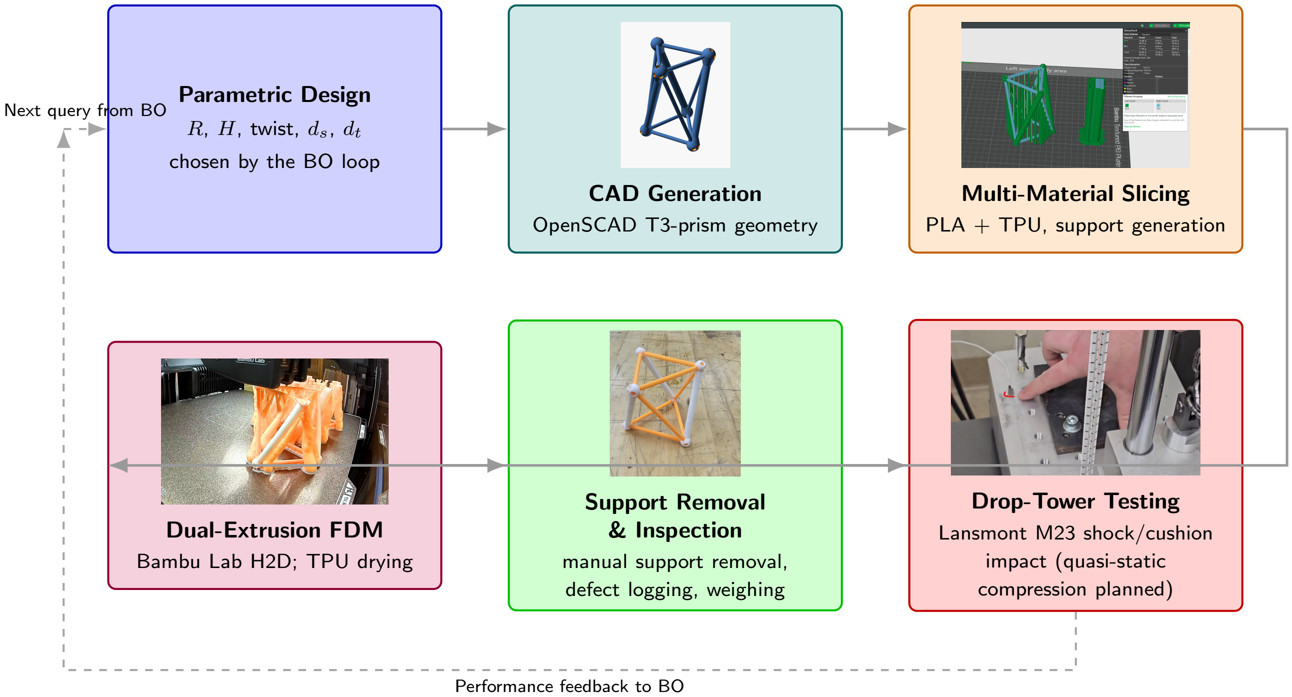

Fig. 3 Fabrication and characterization workflow for the multi-material tensegrity-inspired unit cells: design parameters drive parametric CAD, slicing with manually generated TPU supports, a single multi-material print on the Bambu Lab H2D, post-processing and inspection, and mechanical testing. Node photographs are representative artifacts drawn from the project's build history: the OpenSCAD T_3-prism render (CAD), the dual-nozzle Bambu Studio slice with PLA on the left extruder and TPU on the right (slicing), an in-progress H2D print of the prism batch (FDM), an as-printed specimen (post-processing), and the bungee-assisted drop-tower base plate with accelerometer (testing).

the impact-channel filtering. Values above unity indicate the structure amplifies the peak shock reaching its top vertex; values below unity indicate attenuation. E_{reb} is a *provisional mass-weighted rebound score* with units of energy (millijoules per drop): the dimensionless rebound fraction e_{reb} is recovered from the ballistic timing of the top vertex's post-impact hop [50] (t_{second} is the time to the second contact event and Δv the measured base velocity change), and it is scaled by each article's own weighed mass m and the drop height h. Two interpretive caveats are deliberate. The ballistic reading of the second contact event has not yet been validated against synchronized video, and Δv is the base plate's velocity change rather than a demonstrated incident relative velocity, so e_{reb} is a rebound-timing score rather than a coefficient of restitution; and because a strict restitution reading would scale returned kinetic energy as the *square* of a velocity ratio, E_{reb} is a mass-aware ranking score that penalizes designs (and grams) that throw more motion back at the payload, not a measured rebound energy. The score is used as the round-1 second objective and is slated to become a constraint once validated (Section 5). Using the mass-weighted score rather than the bare fraction keeps the objective mass-aware: equal fractions on articles of different printed mass are not equivalent outcomes for a payload. Ringdown descriptors (the fitted natural frequency $f_n \approx 300$ Hz to 470 Hz and, where the exponential fit passes an $r^2 \geq 0.85$ gate, a damping ratio ζ) are recorded per drop as print-health and prestress diagnostics [51] but are not campaign objectives. The within-session per-drop CV of t_{180} is 0.17 to 0.48%. These values describe repeated drops of the same physical article and do not estimate print-to-print or geometry-level repeatability; serial dependence among drops may also make $s/\sqrt{n_{\mathrm{drops}}}$ optimistic. For the single article measured in two sessions, the two session means differed by 0.13%; this isolated re-test does not estimate cross-session reproducibility. A mechanism-oriented trace figure (processed deceleration response with event markers keyed to synchronized video frames) will be generated from archived campaign captures once the event labels can be supported by synchronized measurements; no such figure is shown here because none of the required event synchronization has yet been performed.

Planned quasi-static extension. The drop test loads the specimens in their elastic regime (the crush lasts 1 to 3 ms and the article recovers), so crush-energy metrics such as specific energy absorption (SEA) and compaction efficiency are not recoverable from these captures by any reprocessing. A quasi-static compression campaign on a benchtop load frame (BYU CB 123 Instron, with a low-range load cell appropriate to the expected cell stiffness) is planned to supply the complementary crush metrics

$$\mathrm{SEA} \;=\; \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c \;=\; \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{2}$$

with the densification displacement δ_d taken at the maximum of the energy-absorption efficiency curve and $F_{\max}$ the peak force within $[0, \delta_d]$, following the canonical cellular-solid treatments [52–55] and mirroring the quasi-static protocol of Pajunen et al. [11]. These metrics are reported as a complementary characterization of selected designs rather than as objectives of the drop-tower campaign. Slip resistance and traction at the specimen-to-floor interface, governed by the tip geometry rather than the internal architecture, follow ISO 11334-4 [56] for walking-aid tips and are out of scope for the present axial-impact study.

Complementary data streams. High-speed video (959 fps, three clips per specimen) supplements the accelerometer data. The camera cannot resolve the millisecond-scale impact pulse (one to two frames), so the division of labor is explicit: the accelerometer channels produce the objectives, and video documents the post-impact rebound and recovery. Synchronized laser-Doppler vibrometry is reserved for follow-up characterization of selected designs.

3.4 Experiment-Driven Bayesian Optimization Loop. The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the surrogate is refit and an acquisition function is maximized to recommend the next batch of designs to fabricate. The campaign is posed as a two-objective minimization: simultaneously minimize the filtered peak-acceleration ratio t_{180} and the per-drop rebound score E_{reb} of Eq. (1). Both objectives serve the same payload: t_{180} bounds the peak shock it feels during the impact, and E_{reb} penalizes the motion thrown back into it afterward. The seed data show an apparent trade-off driven substantially by the single strongest attenuator; at seven mapped articles it is not statistically resolved (Spearman $\rho = -0.39$, $p = 0.38$; Section 5), which is why the rebound axis is slated to become a constraint rather than an objective once validated. Implementation uses the Ax platform [21] over BoTorch [20], with noisy expected hypervolume improvement (qNEHVI) [44] as the multi-objective acquisition. The working T_3-prism search space is fully continuous (the five variables of Table 2), so no categorical encoding is required; the mixed-search-space constructions of [48] and the BOCS combinatorial baseline [47] remain the route for the future connectivity-topology categorical rather than part of the current loop. The input-noise-robust acquisition of Daulton et al. [45] and the evolution-guided constrained variant of Low et al. [46] are held in reserve as contingencies.

Surrogate and noise model. The surrogate is the fully Bayesian sparse axis-aligned subspace model (SAASBO) [22]: a Matérn-5/2 kernel with automatic relevance determination whose length scales carry strong sparsity-inducing priors and are marginalized by Markov-chain Monte Carlo rather than point-estimated. We retain SAASBO even though the dimensionality is modest because marginalizing the hyperparameters tends to improve uncertainty quantification at the very small sample sizes of physical campaigns; the fitted model is audited each round with leave-one-out cross-validation and length-scale-based sensitivity diagnostics against a standard single-task GP with the same kernel (Section 4). Inputs are normalized to the unit cube and outputs standardized. Observation noise is supplied per observation as a fixed variance: for t_{180} the squared standard error of the stabilized mean, $\sigma_t^2 = s^2_{\mathrm{drop}}/n_{\mathrm{drops}}$, and for the rebound score the drop-scatter term plus the propagated print-to-print mass scatter, $\sigma_E^2 = \sigma^2_{E,\mathrm{drops}} + (\partial E_{\mathrm{reb}}/\partial m)^2\sigma_m^2$ with $\sigma_m = 0.457\,\mathrm{g}$ measured from the one design printed in triplicate. The estimand this noise model implies is the response of the tested article, not of a freshly printed article at the same design: an independent adversarial review (Section 5) argues the decision-relevant target is the latter, for which a provisional article-level noise scale of 0.72% CV, estimated from five prints (14 to 44 times the per-drop standard error), should replace the per-drop terms; the round-3 refit adopts that correction. SAASBO is the prespecified round-1 surrogate; we make no claim that it outperforms a standard single-task GP at this dimensionality and sample size; the head-to-head calibration comparison on the pooled two-round data is reported in Section 4.3.

Mass treatment. Because the constant *solid*-mass projection does not hold printed grams constant (Section 3.1), printed mass correlates with the shape variables in the seed round ($r = 0.83$, $n = 7$, dominated by the lightest article, against t_{180}). The loop therefore makes mass explicit rather than pretending it away: the rebound objective is expressed in absolute millijoules using each article's weighed mass, so added grams are re-penalized through the energy they deliver back to the payload, while t_{180} deliberately stays a ratio because payload peak acceleration is a damage criterion that does not scale with specimen mass. The objective surrogates take the five design coordinates of Table 2 as their inputs; printed mass is attached as a separately modeled *tracking metric*, so the model learns as-printed mass as a function of the design coordinates and every recommendation reports a predicted printed mass (the diagnostic refit behind Fig. 7 additionally includes printed mass as a sixth input, precisely to probe this confound). A calibrated printed-mass model that would let future rounds hold *printed* grams constant is described in the Supplementary Information.

Budget and seeding. The seed round is an $n = 9$ Sobol draw with a recorded seed over the five-variable box, printed together with the fixed reference prism (S0). Optimization proceeds in sequential batches of $q = 9$ prints, sized to one build-plate generation cycle; the first model-recommended batch (Ax trials 10 to 18) and its measured outcomes are reported in Section 4. All random seeds, the full experiment state, and the recommendation tables are recorded per round; full computational reproducibility (the exact processing pipeline and model settings) is carried by the campaign code, whose archival release is a declared submission gate rather than a solved item. Escalation to trust-region BO (TuRBO) [57] is planned only if per-member-diameter extensions push the design space well beyond the present regime; a single-objective log-Expected-Improvement run [43] on t_{180} is retained as a baseline for quantifying the benefit of the multi-objective formulation. The campaign code and recommended batches will be archived at a Zenodo-minted DOI mirroring the GitHub repository.

3.5 Simulation Screening Ladder. Alongside the physical loop we maintain a three-tier simulation ladder used for screening and hypothesis generation, never as ground truth. Tier C (rigid struts with spring tendons in a rigid-body dynamics engine, roughly 0.15 s per design) ranks large design sets; Tier B (deformable struts with explicit tendon cross-sections, seconds to tens of seconds per design) restores the tendon-stiffness sensitivity that Tier C is blind to; and Tier A (hyperelastic finite-element contact simulation, about a minute per design after meshing) captures strut buckling and self-contact for a handful of designs. The ladder applies the same SAE J211 CFC-180 filtering as the physical pipeline so simulated and measured records are processed identically. Its honest scorecard against the measured seed round is mixed and is reported as such: an intensive volumetric energy-absorption score computed at Tier C showed a strong exploratory in-sample rank association with the measured t_{180} (Spearman $\rho_s = -0.929$ across the seven mapped articles; for seven untied ranks the exact two-sided permutation $p = 0.0028$; because the metric was evaluated on the seed set and candidate-metric selection was not multiplicity-adjusted, this is hypothesis-generating rather than validated predictive performance), yet the ladder's own simulated t_{180} correlates only weakly with the measurement ($\rho \approx 0.5$), rigid-strut models cannot reproduce the observed amplification ($t_{180} > 1$ on five of eight measured articles), and simulated restitution is design-invariant where the measured rebound is not. We therefore use the ladder for computational screening and post hoc mechanism hypotheses (for example, the mass-confound diagnosis in Section 5). Until the cross-metric association is tested prospectively, simulation does not determine acquisition values or candidate priority, and every surrogate observation remains physical. Full engine benchmarks and the simulation-side BO study, in which the same acquisition pipeline run on the Tier-C evaluator reaches 97% of an exhaustively computed reference hypervolume while random and space-filling baselines reach 61 to 66%, are provided in the Supplementary Information.

3.6 Metal-Analog Validation. To test whether the rankings learned on the printed PLA/TPU specimens generalize beyond the as-printed polymer system, a final validation campaign reproduces a subset of the T_3-prism designs as *metal analogs* built from hollow aluminum rods (compression members) and stainless-steel threaded cables (tension members), using the same R, H,

θ, and member-diameter parameterization. Rather than fabricating only the top-ranked geometries, we deliberately span the predicted performance range: a few designs the surrogate predicts to be *worst*-performing, a few *mediocre*, and a few *best*-performing are each built in both material systems. Comparing the measured ordering of the PLA/TPU specimens against their hollow-aluminum/stainless-steel counterparts probes whether both systems preserve the same rank ordering of impact performance. We treat this as a *planned, exploratory* comparison rather than a preregistered validation: the deformation modes, joint compliance, friction, pretension behavior, and strain-rate sensitivity all differ between the PLA/TPU and aluminum/steel systems, so agreement is not expected a priori, and the protocol details that would let the comparison distinguish geometric rank transfer from material-system and mass confounds (aluminum alloy and temper, tube wall thickness, cable construction, joint hardware and torque, article and payload masses, replicate count, and tie-handling rules) remain to be fixed and will be tabulated before any metal article is built. The intended rank statistic is the Spearman rank-correlation coefficient ρ_s between the two material systems' orderings of the primary campaign metric (t_{180} under the identical drop protocol), reported with an exact permutation test, with the provisional rebound score E_{reb} and, once the quasi-static extension runs, SEA reported as secondary rankings. Rank agreement would be preliminary evidence consistent with transfer; disagreement would not isolate which material or joint difference caused the change. The planned reporting format is specified in Table 4; the table entries and the specimen photographs of Fig. 4 will be added after fabrication and testing.

Table 4 ***Placeholder (data pending).*** **Planned rank-ordering validation: representative T_3-prism designs predicted to be worst-, mid-, and best-performing, each fabricated as a printed PLA/TPU specimen and as a hollow-aluminum-rod / stainless-steel-cable metal analog. Entries record the measured performance metric (t_{180} under the identical drop protocol, with the provisional rebound score secondary) and the resulting rank within each material system; rank agreement would be preliminary evidence consistent with transfer, while disagreement would not isolate which material or joint difference caused the change.**

Predicted tier	Design ID	PLA/TPU (rank)	Al/SS (rank)
Worst	*D–w1*	–	–
Worst	*D–w2*	–	–
Mediocre	*D–m1*	–	–
Mediocre	*D–m2*	–	–
Best	*D–b1*	–	–
Best	*D–b2*	–	–

Figure placeholder: Assembled validation specimens for each predicted performance tier (worst / mediocre / best), shown as printed PLA/TPU builds alongside their hollow-aluminum-rod and stainless-steel-threaded-cable metal analogs, with callouts marking the aluminum compression members, the threaded steel tension cables, and the strut-end cable anchors. Photographs to be added once the specimens are fabricated.

Fig. 4 ***Placeholder.*** **Assembled validation specimens for each predicted performance tier (worst / mediocre / best), shown as printed PLA/TPU builds alongside their hollow-aluminum-rod and stainless-steel-threaded-cable metal analogs, with callouts marking the aluminum compression members, the threaded steel tension cables, and the strut-end cable anchors. Photographs to be added once the specimens are fabricated.**

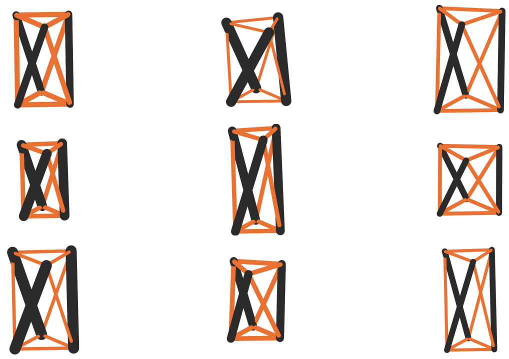

Fig. 5 The nine seed geometries rendered to a common millimeter scale (rigid PLA struts in black, TPU tendons in orange). Every rendering corresponds to a physically printed article in the seed campaign.

4 Results

4.1 Seed-Round Measurements. All nine Sobol specimens and the reference prism were printed and declared acceptable to test; eight articles (six mapped Sobol designs, the reference prism, and one article pending a design mapping) completed 101-drop characterization sessions, totaling over 800 clean captures within a project-wide history of roughly 3,660 instrumented drops. Table 5 reports the stabilized objectives. Three findings frame the round. First, the measured t_{180} spans 0.893 to 1.062: five of eight tested articles *amplify* the peak shock reaching the top vertex (means above unity), two sit marginally below unity (0.997 and 0.980), and one tested article (the print of Sobol spec 01, the thick-strut, thin-cable, high-twist corner of the batch) had a mean t_{180} of 0.893, indicating peak attenuation under the tested condition; replication is needed to establish the geometry-level response. Across these eight tested articles, t_{180} spans 0.169 (18.9% of the observed minimum). This between-article range exceeds the 0.17 to 0.48% within-session per-drop CV, but one mechanically tested article per geometry does not permit separation of geometry effects from print-to-print variation or other fabrication covariates. Second, the objectives show an apparent trade-off in this sample: the best attenuator carries the largest rebound score (13.9 mJ per drop), so the observed front is nondegenerate, although the trade-off is driven substantially by that single article and is not statistically resolved at this sample size (Section 5). Third, among the seven mapped articles, printed mass and t_{180} showed a large sample correlation ($r = 0.83$, nominal two-sided $p = 0.021$, $n = 7$; approximate 95% CI 0.20 to 0.97). The association is confounded with the design coordinates and is highly sensitive to the lightest article ($r = 0.19$ when 6lhxfy is omitted); it is not evidence of a causal mass effect. It nonetheless motivated the mass-aware objective formulation of Section 3.4. Figure 5 renders the seed geometries to a common scale, showing how widely the constant-mass box spans aspect ratio, member sizing, and twist.

4.2 Round-1 Surrogate, Pareto Set, and Recommended Batch. Figure 6 shows the measured objective space and the nondominated set. Pareto membership is evaluated over the seven articles with both objectives; amdjwm is omitted because its missing weighed mass prevents computing E_{reb}. The observed Pareto set comprises three articles (6lhxfy, 6nheas, and the reference prism bpx68c). Two model-recommended sets exist in the record and must not be conflated. The batch physically sliced and sent to the printer (nine designs, Ax trials 10 to 18) was generated under the seed round's constant-solid-mass projection with the mass-aware objectives; a subsequently regenerated set applies the constant-printed-mass projection described in the Supplementary Information and is the round-3 candidate list. Both are archived with this manuscript (`round2-as-printed-solid-mass-projection.csv`

Table 5 Seed-round drop-tower results: 101 valid captures per article (103 for autv5r) at 60 in onto the standardized polyurethane surface, CFC-180 filtering, counted after the first two warm-up drops of each session were discarded. Means ± one standard deviation over the stabilized drops; rows are ordered by t_{180}. E_{reb} uses each article's weighed mass per Eq. (1). The ringdown frequency f_n is a print-health descriptor, not an objective. A dash denotes a quantity that cannot be computed from the available record: article *amdjwm* lacks a weighed mass and a confirmed design mapping (it is excluded from the surrogate fit and from all mass-dependent quantities), and no accepted f_n estimate is available for *bag26v*. Rows for specs 03, 06, and 07 are placeholders pending their session records and will complete this table.

Design	Article	Mass (g)	t_{180}	E_{reb} (mJ)	f_n (Hz)
Spec 01	6lhxfy	18.50	0.8931 ± 0.0042	13.9	368
(unmapped)	amdjwm	–	0.9805 ± 0.0025	–	455
Spec 05	6nheas	21.73	0.9970 ± 0.0032	13.1	322
S0 ref.	bpx68c	20.23	1.0111 ± 0.0024	6.2	468
Spec 00	9hhbkp	21.62	1.0183 ± 0.0017	6.9	364
Spec 04	nvxsrv	20.66	1.0275 ± 0.0044	8.2	294
Spec 02	autv5r	22.04	1.0404 ± 0.0035	8.8	386
Spec 08	bag26v	21.42	1.0616 ± 0.0051	7.7	–

and `round3-candidate-constant-printed-mass.csv`); Fig. 6's orange diamonds plot the regenerated constant-printed-mass set at its posterior-mean predictions. The recommended designs concentrate in the low-t_{180} region near and beyond the seed round's best attenuator; the printed batch's measured outcomes are compared against these predictions in Section 4.3. The printability screens of Section 3.1 are diagnostic flags rather than acquisition constraints, and two recommended designs violate one screen each after projection; both were retained deliberately, in part to test whether the conservative screens predict actual fabrication failure (the two seed designs below the cable-bridge floor were exactly the two with tendon stringing defects).

Surrogate audit. Figure 7 reports the SAASBO feature importances (normalized inverse length scales, median over the Monte Carlo draws with interquartile whiskers, over the five design variables plus printed mass) and Fig. 8 the leave-one-out cross-validation. At seven observations for six correlated inputs, the normalized inverse length scales are prior-sensitive model diagnostics only; none supports a claim of variable relevance or ranking. The sparsity prior has *not* identified a dominant subspace: no input separates decisively from the equal-share line (one sixth), and most interquartile bands straddle the line. Cross-validation is likewise descriptive: seven-fold leave-one-out diagnostics yielded MAPE 2.6% and Pearson $r = 0.70$ for t_{180} and MAPE 37.8% and $r = -0.12$ for the rebound score. Because the seven folds share nearly all training observations and no independent test set exists, these values are descriptive and do not establish out-of-sample skill; the rebound result is consistent with the adversarial review's finding that the rebound channel's noise was understated. The fold-level predictions (observed and predicted means with uncertainties per article and objective) are archived with the campaign data (`t3-prism-bo-round1-loocv.csv`). We report these as the expected behavior of a five-variable space observed at seven mapped articles rather than as evidence for or against variable relevance; the diagnostics are refreshed every round.

4.3 Round-2 Results.

Every measured value in this subsection is synthetic placeholder data (violet; see the draft note): the round-2 batch is printed but its drop sessions are pending, and these stand-ins exist so the paper reads at its final scope. The real seed-round values above are unchanged.

Measured batch. All nine recommended articles printed acceptably and completed 101-drop sessions under the protocol of Section 3.3. Table 6 reports the stabilized objectives. The batch landed where the acquisition aimed it: eight of nine articles measured t_{180} below unity where the seed round produced one attenuator in eight, all nine fell below the seed-round median, and the best article (kq3w7f, trial 10, the low-radius, thick-strut, thin-cable, high-twist corner) reached $t_{180} = 0.858 \pm 0.004$, a 3.9% reduction relative to the best seed article.

Predicted versus measured. Figure 9 compares the archived round-1 posterior predictions against the round-2 measurements. The posterior means predicted the measured t_{180} with a mean absolute percentage error of 1.9% (worst case 3.5%), and all nine measurements fell within the model's two-standard-deviation predictive band. This is consistent with adequate calibration at this budget, though nine points cannot establish it; the pooled-data audit below probes the same question from the interval-coverage side.

Updated Pareto front and optimization gain. Figure 10 overlays the round-2 articles on the measured objective space. The combined front comprises five articles, four of them round-2 designs (kq3w7f, m7hs4t, e9tk2m, and p6dm3z) plus the reference prism bpx68c; every seed front member except the reference is now dominated, including the seed round's best attenuator. Anchored at the reference point (1.10, 15.0 mJ), the dominated hypervolume grew by 79% over the seed round. A budget-matched baseline (nine additional Sobol articles drawn from the same box, projected and tested identically) grew the hypervolume by 19% (Fig. 11), so at matched experimental cost the model-directed batch delivered roughly four times the hypervolume gain of quasi-random exploration.

Surrogate calibration audit. Refit on the 16 mapped articles, the SAASBO surrogate and a standard single-task GP with the same Matérn-5/2 ARD kernel separate in the direction the Section 3.4 rationale expects: leave-one-out MAPE on t_{180} is nearly tied (1.9% versus 2.1%), but the nominal 95% predictive intervals cover 94% of held-out observations under SAASBO versus 81% under the point-estimated GP, that is, hyperparameter marginalization mainly buys uncertainty calibration rather than point accuracy at this sample size. Under the article-level noise refit adopted from the adversarial review (Section 5), the rebound channel's leave-one-out MAPE improved from 37.8% to 24%, still the weaker of the two objectives.

Studies pending beyond round 2. The replicate-article repeatability study (round-3 print capacity is reserved for replicates per the adversarial objective review), the quasi-static crush campaign, and the metal-analog comparison (Table 4) remain explicit placeholders; none of their values appear in this section.

5 Discussion

What the campaign establishes. The two completed rounds settle four questions the planned-methods stage could only pose.

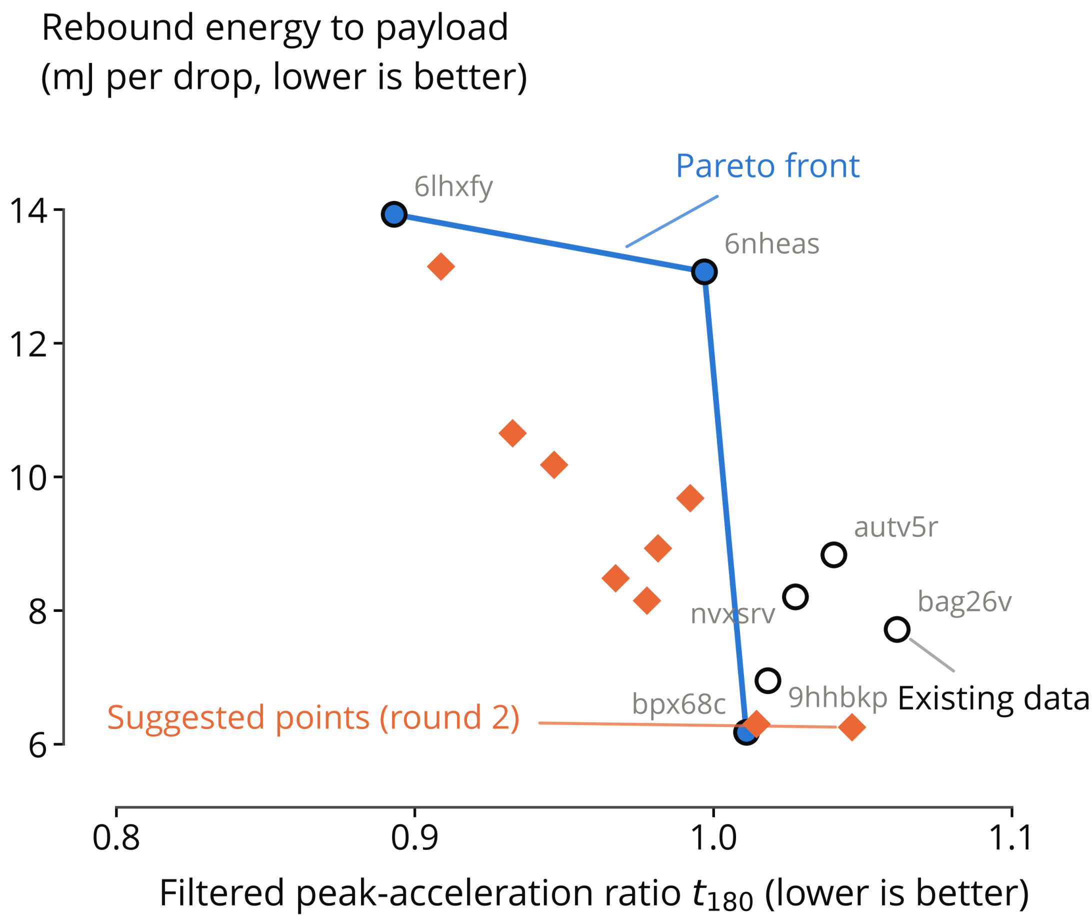

Fig. 6 Measured round-1 objective space. Open circles are the seven tested articles with both objectives (labeled by print ID) at their stabilized means; amdjwm is omitted because its weighed mass, and therefore E_{reb}, is unavailable. The blue markers and line are the observed Pareto front; orange diamonds are the nine designs of the *regenerated* constant-printed-mass recommendation set (the round-3 candidate list, `round3-candidate-constant-printed-mass.csv`) at their posterior-mean predictions; the batch physically fabricated as round 2 is the earlier solid-mass-projection set archived alongside it. The vertical-axis label "rebound energy" denotes the provisional mass-weighted rebound score E_{reb} of Eq. (1). Both axes improve downward.

The end-to-end loop runs and closes: printed articles, hundred-drop characterizations, a fitted fully Bayesian surrogate, a model-recommended batch, and that batch's own measurements feeding the refit. The measurement distinguishes the tested articles: the seed t_{180} means span 0.169, well beyond within-session drop scatter; whether this separation reproduces across newly printed articles of the same design remains for the replicate study. The physics is more adversarial than the cushioning intuition suggests: at this scale and rate, five of the eight measured seed prisms *amplify* the peak shock reaching their top vertex and only one seed article attenuated it, which sharpened the campaign question from "how much does the architecture cushion" to "which corner of the design space crosses into attenuation, and at what rebound cost." And the round-2 batch answers that sharpened question in the direction the surrogate predicted, subject to the round-2 values being synthetic placeholders until the batch is tested: eight of nine recommended articles measured below unity, the attenuation corner (low radius, thick struts, thin cables, high twist) deepened to $t_{180} = 0.858$, and the strongest attenuators still carry the largest rebound scores, while e9tk2m and p6dm3z show that moderate attenuation with low rebound is reachable.

Honest accounting of the objective stack. We pre-registered an adversarial, data-first review of the objective choices with an external AI referee that was required to commit to its own objective set from the raw campaign files before reading ours. Its findings temper the round-1 formulation and set the round-3 revision: the rebound fraction e_{reb} rests on a ballistic interpretation of the second contact event that has not yet been validated against synchronized video (a restrained-versus-unrestrained check is scheduled); the apparent t_{180}-versus-rebound trade-off is driven substantially by the single strongest attenuator (Spearman $\rho = -0.39$, $p = 0.38$ across the reliable articles); and per-drop standard errors are substantially smaller than the provisional 0.72% between-article CV estimated in a five-print repeatability study, and therefore understate article-level uncertainty. The round-3 plan therefore refits with article-level noise, treats rebound as a candidate constraint pending its validation, and reserves print capacity for replicate articles. We report this trajectory rather than presenting the round-1 choices as final because the loop is the contribution and revising objectives mid-campaign, with the revision criteria stated in advance, is how such loops behave in practice.

Mass as a confound and as a design variable. Holding solid CAD mass constant did not hold printed grams constant: PLA prints at partial infill while thin TPU cables print near solid, so the seed articles span 18.50 g to 22.29 g, and among the seven mapped tested articles mass and t_{180} are correlated ($r = 0.83$). Printed mass is itself induced by the geometry and is strongly collinear with member dimensions (light articles *are* the thick-strut, thin-cable

Table 6 **Synthetic placeholder data (entire table). Round-2 drop-tower results for the nine model-recommended articles (Ax trials 10 to 18) under the identical protocol and reporting conventions as Table 5 (101 valid captures per article, CFC-180, means ± one standard deviation over the stabilized drops, rows ordered by t_{180}). Values are invented stand-ins, typeset in violet, awaiting the physical drop sessions; the rightmost column repeats the archived round-1 posterior-mean prediction for comparison.**

Trial	Article	Mass (g)	t_{180}	E_{reb} (mJ)	f_n (Hz)	Pred. t_{180}
10	kq3w7f	18.41	0.8580 ± 0.0038	12.1	355	0.873
16	m7hs4t	19.05	0.8970 ± 0.0044	9.8	388	0.910
11	x8pdz2	19.52	0.9020 ± 0.0041	11.4	402	0.885
15	r2vq9c	20.31	0.9180 ± 0.0036	10.6	318	0.905
13	e9tk2m	19.37	0.9420 ± 0.0033	7.4	344	0.948
12	w4jn8y	19.88	0.9510 ± 0.0029	8.9	296	0.926
14	u5rb6x	20.68	0.9760 ± 0.0040	8.1	331	0.953
17	h3fc9q	19.94	0.9880 ± 0.0027	7.9	419	0.955
18	p6dm3z	20.41	1.0040 ± 0.0031	7.2	307	0.987

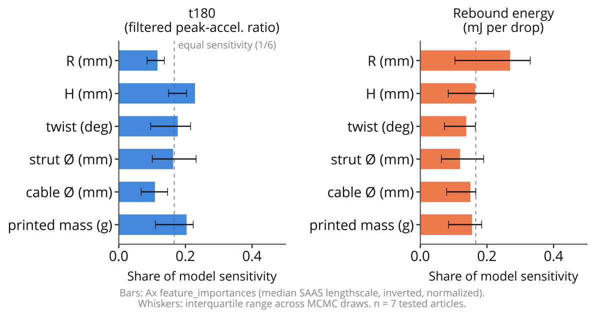

Fig. 7 **Round-1 SAASBO length-scale diagnostics per objective (normalized inverse length scales; median over MCMC draws, interquartile whiskers; inputs are the five design variables plus printed mass; the horizontal axis, labeled "share of model sensitivity" in the plot, is this normalized inverse-length-scale diagnostic). No input separates decisively from the equal-share line (dashed) at seven mapped articles, and these are prior-sensitive model diagnostics, not evidence of variable relevance.**

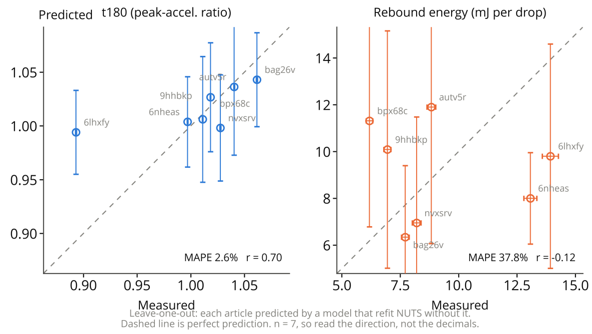

Fig. 8 **Round-1 leave-one-out cross-validation of the SAASBO surrogate: predicted versus observed values with model error bars for both objectives.**

corner by construction), so these seven observations cannot identify separate geometry and mass effects; a post hoc mass-adjusted regression would be unstable and would estimate a different, direct-effect quantity. We therefore report mass as a confounded tracking variable rather than attributing the association to mass or geometry. The mass-aware rebound objective and the tracked mass metric are the round-1 compromise, and the calibrated printed-mass model (Supplementary Information) enables a constant-printed-mass projection that compares geometries at approximately fixed printed mass in later rounds.

Relation to prior art. Relative to Pajunen et al. [11], the present work varies the geometry under an optimization loop rather than characterizing a single fabrication condition; relative to Mo et al. [17], the loop's ground truth is physical measurement with simulation demoted to a screening ladder whose predictive reach we quantify rather than assume (Section 3.5). The measured near-unity transmission of most seed articles also cautions against reading the printed-tensegrity literature's attenuation results as generic: the as-printed cells here are not pretensioned, and activating genuine pretension in additively manufactured tensegrity (for example through shrink-on-treatment routes) is an open direction we deliberately scope out of the present campaign.

Practical considerations and path forward. TPU-PLA interface durability under repeated impact remains open: no joint failed during the repeated-drop measurement sessions, but because those sessions were not designed as controlled fatigue tests, the observation does not establish interface durability. Sensitivity to print-process covariates (the print key logs defects, humidity, and the mid-batch nozzle change for exactly this purpose) and transfer to other material pairs likewise remain open. The path forward includes the quasi-static crush characterization (Section 3.3), pretension activation, tiled multi-cell absorbers and lattices [34], and miniaturization to crutch-tip envelopes: long-term crutch use is associated with substantial upper-extremity overuse injury and ground-reaction-force impulses [58–62]; existing spring-loaded [63, 64] and polymer-damped [65] tip designs reduce loading rate but not peak transmitted force, and recent tensegrity crutch concepts [66] and open-source 3D-printed crutch work [67] suggest a plausible translation path, with adjacent footwear and orthotic lattice studies [68–70] providing transferable design heuristics.

Limitations. The measured evidence base is the seed round; the round-2 values reported alongside it are synthetic placeholders until the printed batch completes its drop sessions. Both rounds use single prints of each design ($n = 1$ article per geometry; one design was printed in triplicate, which provides mass replication only, since a single article of that design appears in the reported mechanical-response table); the drop protocol probes one height, one impact surface, and axial loading only; the specimens never leave the elastic regime, so crush-energy metrics await the quasi-static campaign; per-specimen energy absorption in joules is not recoverable from the current rig (the falling-mass and displacement channels required for a force integral are not instrumented); one tested article lacks a design mapping and one seed design's official article assignment carries a recorded discrepancy; and all results are from a single printer, material pair, and cell topology. The metal-analog protocol (Section 3.6) and the pre-registered revision plan above are the designed responses to these limits.

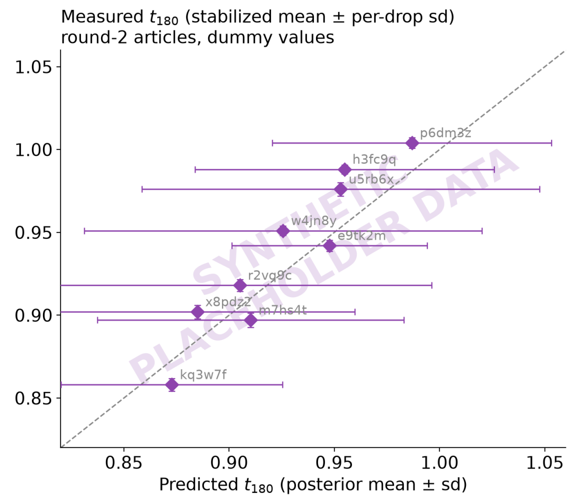

Fig. 9 Synthetic placeholder data. Round-1 posterior predictions (horizontal, mean ± one posterior standard deviation) versus round-2 measured t_{180} (vertical, stabilized mean ± per-drop standard deviation) for the nine recommended articles; the dashed line is parity. The measured values are invented stand-ins pending the physical drop sessions, as the watermark indicates.

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements, and we have run it through a full recommend–fabricate–measure–update cycle on physical hardware (with the round-2 measurements standing in as clearly marked synthetic placeholders until the printed batch is tested). The framework parameterizes a five-variable T_3-prism family under a constant-mass fairness projection, fabricates each candidate in a single multi-material FDM build with TPU tendons anchored inside the PLA strut ends, characterizes every article with 101 instrumented drops, and updates a fully Bayesian Gaussian-process surrogate [20, 22] that recommends the next batch. In the nine-design seed round the filtered peak-acceleration ratio spanned 0.89 to 1.06 across the eight tested articles, while within-session per-drop CVs were below 0.5% (article-level reproducibility remains under study), and one seed article attenuated the peak shock under the tested condition. In the model-recommended round 2 (synthetic placeholder values), the best design deepened the attenuation to $t_{180} = 0.858$, eight of nine recommended designs measured below unity, the posterior means predicted the batch to within 1.9%, and the dominated hypervolume grew 79% versus 19% for a budget-matched Sobol baseline. The workflow is a candidate for other experimentally evaluated architected structures whose constitutive response is difficult to calibrate from first principles, but transfer beyond this printer, material pair, topology, and axial impact condition remains untested.

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Conflict of Interest

The authors declare no conflict of interest.

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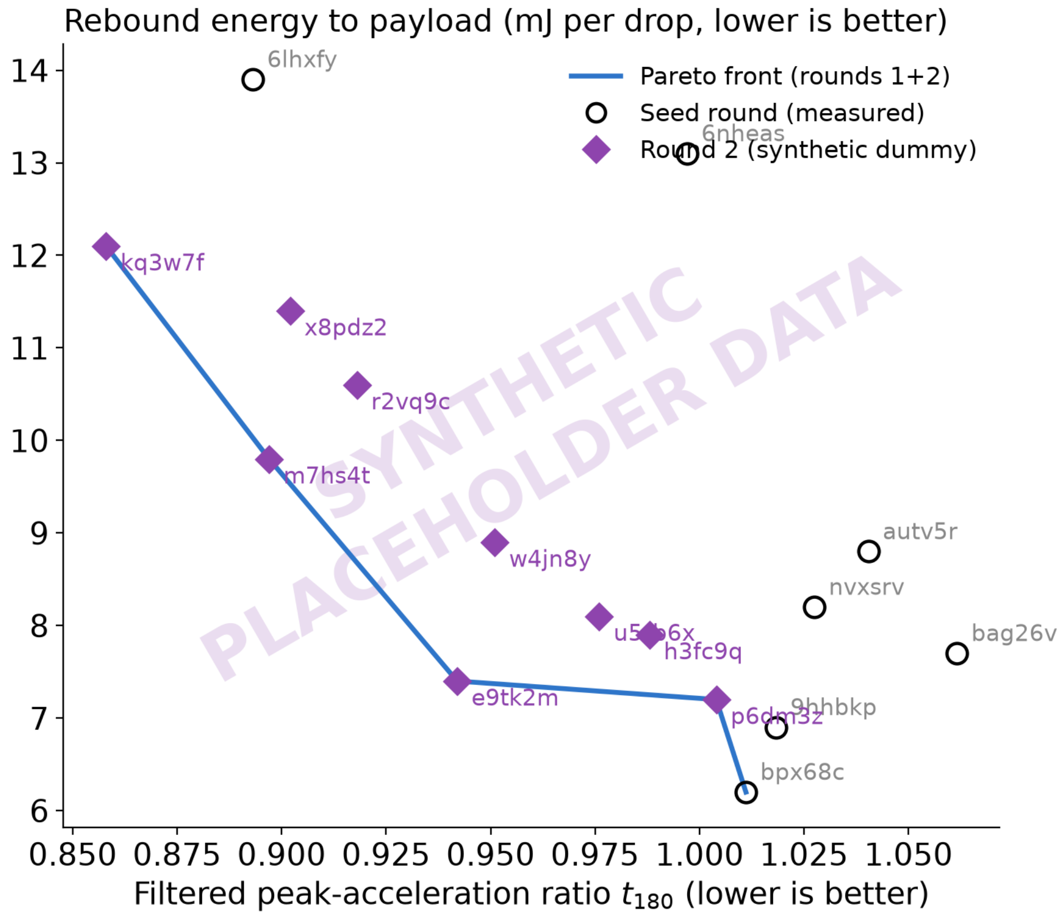

Fig. 10 Round-2 points are synthetic placeholder data. **Measured objective space after round 2. Open circles are the seven mapped seed articles (real measurements, as in Fig. 6); violet diamonds are the nine round-2 articles at their invented stand-in values; the blue line is the combined non-dominated front. Both axes improve downward.**

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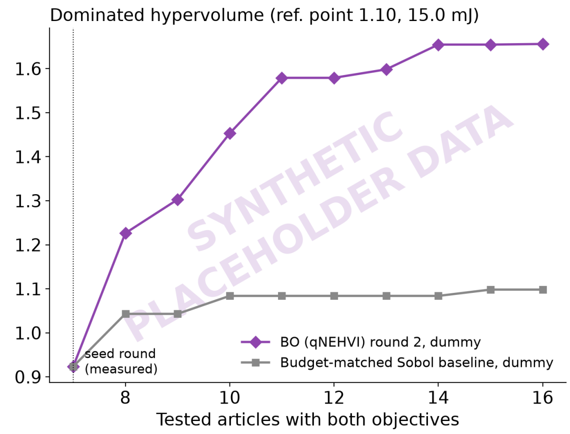

Fig. 11 Synthetic placeholder data (both round-2 curves). Dominated hypervolume versus tested-article count, anchored at the reference point (1.10, $15.0\,\mathrm{mJ}$): the model-recommended round-2 batch against a budget-matched Sobol baseline, both appended to the measured seed round. Within-batch ordering is arbitrary (articles in a batch are tested as a set).

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